

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Hydra Shield: An Intelligent Flood Prediction and Early Warning System (Rising Waters)
Maximum marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, Flask	Provides a responsive and user-friendly interface for entering weather parameters and displaying flood prediction results.
2	Backend / Server-Side	Python, Flask, Scikit-learn	Handles application logic, processes user inputs, loads the trained machine learning model, and generates real-time flood risk predictions.
3	Database / Data Storage	CSV Dataset, Pickle Model	Stores historical flood data and the trained machine learning model for prediction.
4	Machine Learning	Decision Tree, Random Forest, K-Nearest Neighbors (KNN), XGBoost	Trains and evaluates multiple machine learning models to achieve accurate flood prediction.
5	Version Control & CI/CD	Git, GitHub	Manages source code, tracks project changes, and supports collaboration.
6	Libraries / Other Tools	Pandas, NumPy, Matplotlib, Joblib	Supports data preprocessing, numerical computation, visualization, model serialization, and deployment.